

CONSTRUCTION OS

PRD - Platform UI/UX Architecture & Premium Design System

Implementation-ready Product Requirements Document for Product, UX/UI, Frontend, Mobile, Platform, Security, QA, DevOps and Enterprise Operations Teams

Product purpose

Define one premium, permission-aware and configuration-driven interface architecture that makes every ConstructionOS module easy to populate, understand and operate.

Document: PRD_Platform_UI_UX_Architecture_Premium_Design_System

Version: 1.0

Status: Implementation-ready functional and technical specification

Platform scope: Internal workspace, executive workspace, mobile, portals, BIM and Platform Administration

Visual direction: Light mode, dark-grey typography, restrained gold/company accents and refined motion

Date: 2026-08-06

ConstructionOS Confidential

Document Control

Field	Value
Document	PRD_Platform_UI_UX_Architecture_Premium_Design_System
Version	1.0
Status	Implementation-ready product, interaction and frontend architecture specification
Primary owner	ConstructionOS Product Design and Frontend Platform Architecture
Audience	Product, UX/UI, frontend, mobile, backend, security, QA, DevOps, SRE and module teams
Primary surfaces	Desktop web, tablet, mobile/PWA, Client Portal, Supplier Portal and Platform Administration
Design policy	One shared design system and page-pattern library across all modules
Saving policy	Manual Save plus Universal Draft Mode before formal submission
Configuration policy	Every module/page/section/widget/field/action/report is configuration-aware
Accessibility target	WCAG 2.1 AA baseline with reduced motion and no color-only communication
Date	2026-08-06

Permanent UI rule

Show the user only what matters now. Preserve full enterprise detail behind progressive disclosure. Keep every important action consistent, every status understandable and every record traceable.

Table of Contents

Sections 1-31	Sections 32-62
1. Executive Summary	32. Validation, Errors and Concurrency
2. Product Vision and Business Outcomes	33. Required UI States
3. Success Metrics and Guardrails	34. Empty States
4. Scope, Non-Goals and Assumptions	35. Contextual Help
5. User Groups and Experience Surfaces	36. Responsive Architecture
6. Permanent UX and Product Principles	37. Mobile and Field Experience
7. Experience Architecture	38. Offline Experience
8. Platform Application Shell	39. BIM and Spatial Experience
9. Workspace Context and Scope Switching	40. Client and Supplier Portals
10. Navigation Architecture	41. Platform Configuration UI
11. Role Centers	42. Permission-Aware Interface
12. Work Inbox	43. Design System Foundations
13. Quick Create	44. Visual Tokens: Color, Type, Spacing and Shape
14. Universal Page Taxonomy	45. Buttons, Actions and Confirmation
15. Universal Record Page	46. Drawers, Dialogs and Full-Page Editors
16. Record Relationships and Context Drawers	47. Accessibility and Localization
17. Form Architecture	48. Motion and Loading
18. Universal Draft Mode	49. Navigation Memory and Workspace Restoration

Sections 1-31	Sections 32-62
19. Guided Wizards	50. Component Library
20. Company Setup Center	51. Schema-Driven Forms
21. Project Setup Center	52. Frontend Technical Architecture
22. Import Center	53. Routes, State and Data Fetching
23. Registers and Data Tables	54. Security and Privacy
24. Filters, Views and Personalization	55. Analytics and UX Telemetry
25. Approval Workspace	56. Non-Functional Requirements
26. Transaction and Journal Workspace	57. Testing Strategy
27. Activity, Comments and Timeline	58. Acceptance Criteria
28. Documents UI	59. Delivery Phases
29. Global Search	60. Page and Flow Registry
30. Notifications and Event Center	61. Risks and Open Decisions
31. Status and Lifecycle Design	62. Definition of Done

PART I - Product Direction and Experience Foundations

1. Executive Summary

ConstructionOS is a premium private operating system for construction companies. Its interface must make a very large capability set feel focused, calm and predictable for each user. The platform will not be designed as a collection of independent module interfaces. Every module will use one shared shell, design system, page-pattern library, permission model and configuration framework.

The UI architecture prioritizes guided setup, role-specific work, progressive disclosure, consistent record pages, powerful but controlled data registers, clear lifecycle states and safe transitions. A new company should be able to populate its structure through Setup Centers, templates, imports and provider connections rather than manually discovering every configuration screen.

Permanent product position: complexity belongs in the platform architecture, not in the user interface and not in duplicated module code.

ConstructionOS Experience Architecture

One shared design system across focused role and channel experiences

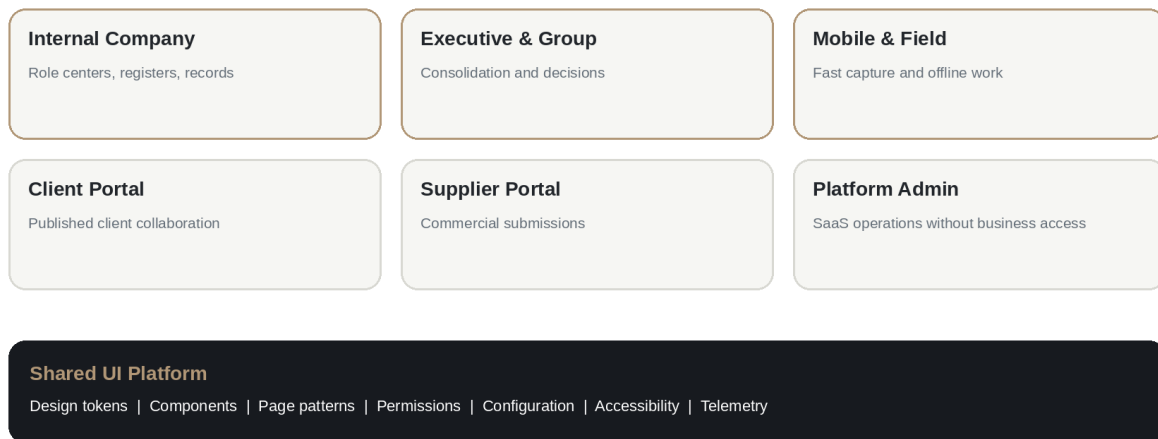


Figure 1 - Shared experience architecture across internal, external and administrative surfaces.

2. Product Vision and Business Outcomes

- Create one recognizable ConstructionOS experience across every module and channel.
- Reduce time required to create a company, project, department, user, supplier, contract or operational record.
- Make enterprise depth available without presenting every field and relationship at once.
- Give each role a purpose-built home and work queue.
- Keep high-risk actions deliberate, traceable and server-confirmed.
- Allow companies to configure terminology, templates and visibility without fragmenting the product.
- Make field workflows fast, touch-friendly and offline-aware.
- Provide a premium visual identity based on clarity, spacing, precision and restrained motion.

ID	Area	Requirement	Priority	Acceptance signal
UI-VIS-001	Consistency	All modules use the shared shell, design tokens, page patterns and component library.	Must	No production module introduces a separate visual language.
UI-VIS-002	Focus	Navigation and dashboards are role-, scope- and subscription-aware.	Must	Users do not see irrelevant modules or actions.
UI-VIS-003	Data entry	Company and Project Setup Centers provide guided completion paths.	Must	Core setup can be completed without navigating unrelated administration pages.

ID	Area	Requirement	Priority	Acceptance signal
UI-VIS-004	Progressive disclosure	Advanced details remain available without dominating the primary workflow.	Must	Primary paths show common and required fields first.
UI-VIS-005	Premium quality	Visual and interaction design is calm, restrained and consistent.	Must	Design review confirms token and motion compliance.

3. Success Metrics and Guardrails

Metric	Target behavior	Guardrail
Time to first usable company	Owner completes minimum company setup in one guided session.	Readiness may not hide mandatory governance gaps.
Time to create common record	Common drafts can be started in seconds through contextual Create.	Quick Create cannot bypass business rules.
Form completion	Users understand required fields and unresolved issues before submission.	No hidden validation only after final click.
Navigation efficiency	Role users reach common work within two navigation decisions.	Do not overload primary navigation.
Error recovery	Drafts and conflicts preserve user work.	Never silently overwrite newer server data.
Mobile completion	Field users can capture assigned work with minimal typing.	Formal release and posting remain controlled online.
Visual consistency	Shared components cover nearly all production screens.	Exceptions require design-system review.
Accessibility	Core flows meet WCAG 2.1 AA.	No status or error depends on color alone.

4. Scope, Non-Goals and Assumptions

4.1 In scope

- Application shell, navigation, context switching and responsive behavior.
- Role Centers, Work Inbox, Quick Create, search and notifications.
- Universal record, form, register, wizard, transaction and approval patterns.
- Company and Project setup, imports, templates and configuration publication.
- Internal workspace, mobile/field, portals, BIM and Platform Administration surfaces.
- Design tokens, component library, schema-driven forms and frontend technical architecture.
- Accessibility, localization, loading, errors, conflicts, offline states and UX telemetry.

4.2 Non-goals

- The PRD does not redefine module business rules or source-of-truth ownership.
- The PRD does not permit UI configuration to override permissions, workflow or immutable lifecycle rules.
- The initial release does not provide arbitrary dashboard drag-and-drop or arbitrary custom modules.
- The initial release does not require a keyboard-shortcut command palette.
- The design system does not replace controlled document templates or BIM viewer rendering engines.

5. User Groups and Experience Surfaces

Surface	Primary users	Experience priority
Internal Company Workspace	Owners, Admins, executives, department and project users	High information density with clear hierarchy and role focus.
Executive and Group Workspace	Parent Owner, CEO, group and shared-service managers	Portfolio visibility, exceptions and decisions rather than operational noise.
Mobile and Field	Site managers, engineers, inspectors, HSE, storekeepers and technicians	Assigned work, evidence capture, scanning, offline state and large controls.
Client Portal	Clients, buyers, client representatives	Published information, simple submissions, documents, handover and support.
Supplier Portal	Suppliers, subcontractors and consultants	RFQs, quotations, deliveries, claims, invoices and compliance submissions.

Surface	Primary users	Experience priority
Platform Administration	Platform operations, billing, support and security roles	Tenant-safe SaaS administration without routine business-data access.

6. Permanent UX and Product Principles

1. Role before module: the interface begins from what the user must accomplish, not from a catalogue of all available data.
2. One visual language: every module must feel like a part of ConstructionOS.
3. Progressive disclosure: common, required and stage-relevant information appears first.
4. Manual Save and controlled drafts: users can preserve work without triggering formal consequences.
5. Server-confirmed high-risk transitions: approval, posting, issue, publication and acceptance are never optimistic.
6. Permission-aware from query to pixel: restricted fields are not fetched and merely hidden.
7. Clear source ownership: projections and related records identify their authoritative module.
8. Mobile is a focused field surface, not a compressed desktop application.
9. Accessibility is part of the component contract, not a later audit-only activity.
10. Motion communicates state and spatial relationships; it never delays work.

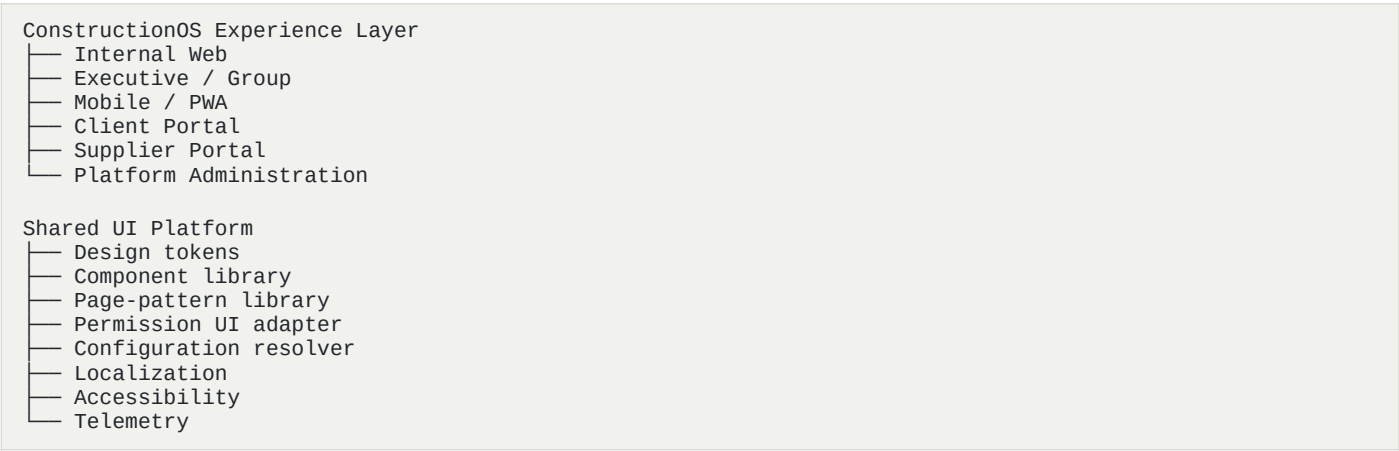
Critical design rule

The interface may simplify presentation, but it must never simplify away ownership, lifecycle, permission, revision, audit or conflict information required for a safe decision.

PART II - Global Shell, Navigation and Role Experience

7. Experience Architecture

All experience applications consume the same design tokens, interaction contracts and authorization results. Portals and mobile use different information density and terminology, but they do not invent parallel component systems.



8. Platform Application Shell

Desktop Application Shell

Persistent context, focused navigation, authoritative workspace, optional supporting context

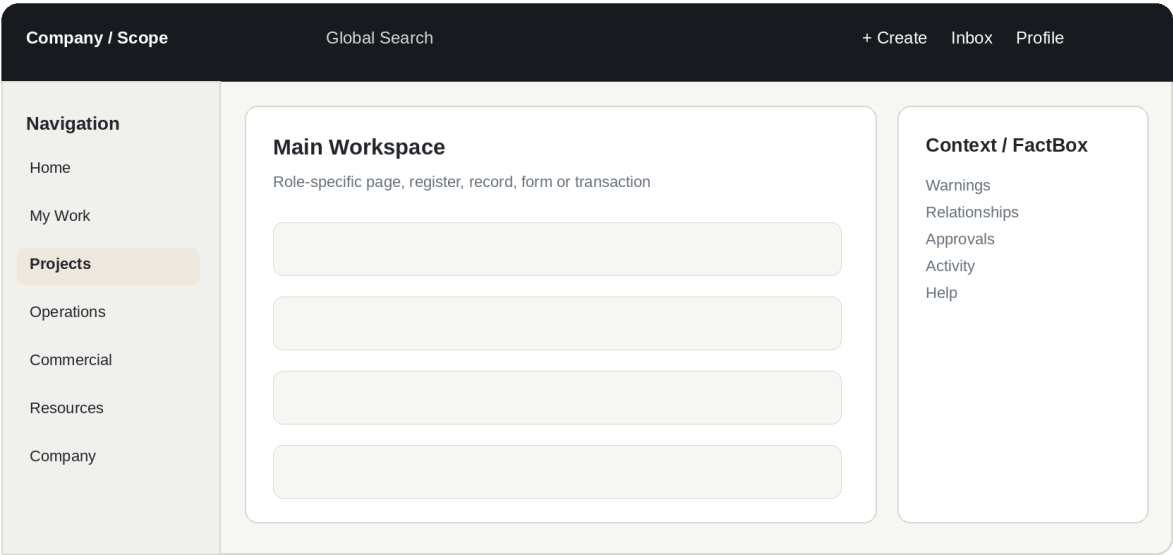


Figure 2 - Desktop shell with persistent context, focused navigation and optional FactBox.

ID	Area	Requirement	Priority	Acceptance signal
UI-SHL-001	Desktop	Desktop uses collapsible left navigation, persistent top bar, main canvas and optional context region.	Must	All internal modules render inside the standard shell.
UI-SHL-002	Tablet	Tablet supports two-column workspaces, slide-over navigation and full-screen complex editors.	Must	No desktop-only workflow blocks tablet use.

ID	Area	Requirement	Priority	Acceptance signal
UI-SHL-003	Mobile	Mobile uses single-column priority layout, bottom navigation and full-screen forms.	Must	Primary mobile flows remain usable at supported widths.
UI-SHL-004	Status bar	Draft, offline, sync, conflict or impersonation status can be displayed persistently.	Must	Critical session state remains visible.
UI-SHL-005	Responsive	Sidebar collapses automatically at configured breakpoints.	Must	Content does not clip or create horizontal application scroll.

9. Workspace Context and Scope Switching

The active business context is a first-class interface element. Users may belong to several companies and Projects and may hold different roles in each. Scope changes must be explicit and must refresh all permission-dependent interface state.

Context element	Required behavior
Active scope label	Display group/company/JV/project hierarchy and current role where useful.
Switch action	Open a searchable list of authorized scopes only.
Refresh	Refresh navigation, Role Center, search, notifications, saved views and actions.
Sensitive confirmation	Repeat company and Project context before posting, approval, export or access change.
Deep links	Resolve the required context before loading the linked record.
Single-company user	Hide unnecessary scope-switcher complexity.

10. Navigation Architecture

Home
My Work
Projects
Operations
Commercial
Resources
Company

Each primary area expands only to modules available under the current role, subscription, scope and Platform Configuration. History remains accessible when a feature is later disabled, but disabled create/actions disappear cleanly.

ID	Area	Requirement	Priority	Acceptance signal
UI-NAV-001	Primary menu	Primary navigation groups modules by work domain rather than a flat module list.	Must	No default sidebar presents all modules as one uninterrupted list.
UI-NAV-002	Visibility	Menu entries resolve from permission, subscription and configuration.	Must	Unauthorized or disabled modules do not appear.
UI-NAV-003	Favorites	Authorized Projects and records may be favorited.	Should	Favorites do not bypass permission revalidation.
UI-NAV-004	Recent	Recently opened records are available by user and scope.	Should	Revoked records disappear immediately.
UI-NAV-005	Deep links	Deep links preserve context and selected tab.	Must	Opening a deep link never widens access.

11. Role Centers

Every user lands on a fixed, purpose-built Role Center. Role Centers are composed from governed zones rather than arbitrary user-built dashboards.

Role Center zone	Purpose
Priority Work	Tasks, decisions, reviews and actions requiring the user.
Critical Exceptions	Overdue, blocked, unsafe, rejected or financially material conditions.
Current Performance	Role-relevant KPIs with governed definitions.
Upcoming Deadlines	Immediate future obligations and scheduled work.
Scope Portfolio	Projects, departments, teams, companies or assets in the role scope.
Recent Records	Recently opened or updated authorized records.
Quick Actions	High-frequency create and navigation actions.
Insights	Permission-safe contextual analytics and reconciliations.

Example Role Center	Primary focus
Company Owner / CEO	Company performance, critical risk, approvals and portfolio.
Project Manager	Work Inbox, milestones, progress, quality, HSE, procurement and team.
Finance	Posting queues, cash, receivables, payables, budget and reconciliation.
HR	Joiners, leave, timesheets, payroll readiness and workforce gaps.
IT	Provisioning, integrations, devices, licences, service desk and security.
Site Operations	Today's work, inspections, issues, deliveries and field capture.

12. Work Inbox

The Work Inbox is the operational center for all actions that require the user. It consolidates actionable items without copying source-module state.

Inbox type	Examples	Primary action
Task	Assigned task, corrective action, meeting action	Open / complete permitted work
Decision	Approval, review, certification, acceptance	Open decision workspace
Reading	Required document or policy	Open and acknowledge
Clarification	Returned submission or information request	Respond
Exception	Conflict, rejected sync, integration failure	Resolve or escalate
Deadline	Expiring document, overdue responsibility	Open source record

ID	Area	Requirement	Priority	Acceptance signal
UI-INB-001	Priority	Inbox orders work using governed priority, due date and risk, not only recency.	Must	Critical assigned items cannot be buried by low-value updates.
UI-INB-002	Source	Every item identifies source module, record and current stage.	Must	Inbox never becomes a competing source of truth.
UI-INB-003	Views	Provide My Priority, Due Today, Overdue, Waiting, Delegated and Completed.	Must	Counts reconcile with source filters.
UI-INB-004	Mobile	Work Inbox appears first on mobile Role Centers.	Must	Field users reach assigned work immediately.

13. Quick Create

The global Create action exposes only authorized, relevant record types. It adapts to the current company, Project, module and device.

Creation mode	Best use
Quick drawer	Small reversible drafts with few fields.
Full-page form	Complex or high-risk records.
Guided wizard	Setup or multi-stage creation.
Template	Repeatable company or Project standards.
Create from source	Task from issue, PO from award, meeting from record.
Scan / QR	Field record creation from location, asset or unit.
Import	High-volume structured data.

PART III - Universal Page and Data Entry Patterns

14. Universal Page Taxonomy

Pattern	Purpose	Examples
Role Center	Priorities, KPIs and exceptions	CEO, Project Manager, IT
Register	Search, filter, bulk action and export	Projects, Employees, Contracts
Split list/detail	Rapid review without losing list context	Tickets, defects, notifications
Record page	Complete authoritative record	Project, Contract, Asset
Wizard	Guided complex creation or setup	Company, employee, integration
Transaction workspace	Line-intensive controlled entry	Journal, invoice, payroll run
Approval workspace	Evidence review and decision	PO approval, handover acceptance
Schedule workspace	Time and resource planning	Calendar, deliveries, shifts
Spatial workspace	Model, map or floor-plan context	BIM, location, units
Configuration workspace	Rules, templates and settings	Permissions, workflows, IT

15. Universal Record Page

Universal Record Page Anatomy

Every major record uses the same structural language

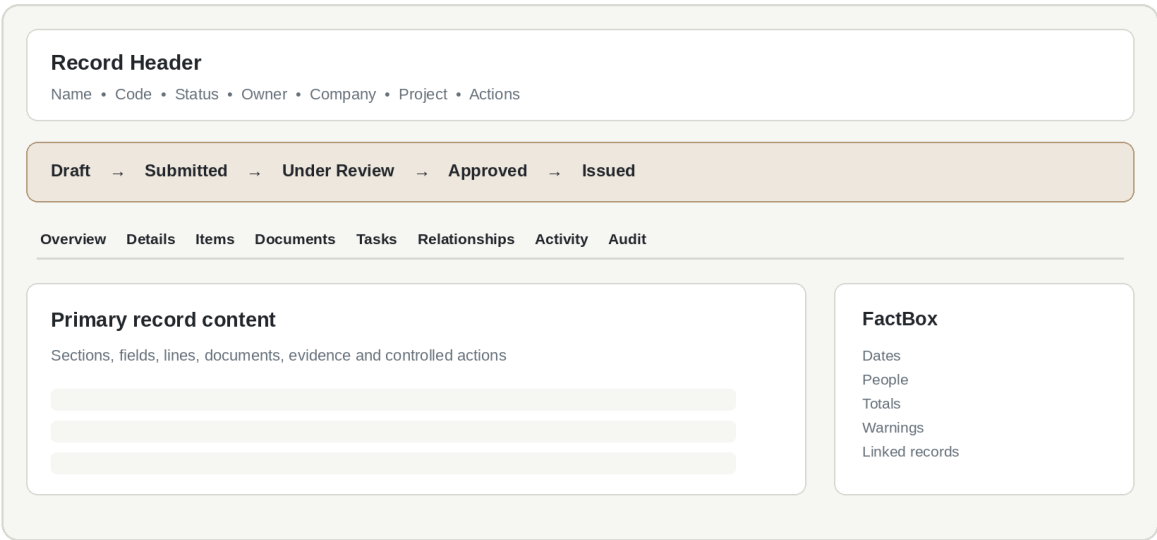


Figure 3 - Standard record anatomy used across all authoritative domains.

Region	Required content
Record Header	Name, code, type, status, owner, scope, revision, update and actions.
Lifecycle Rail	Current state, completed states, blockers, next actor and workflow stage.
Primary Summary	High-value fields and immediate warnings.
Tabs	Overview, Details, Items, Documents, Tasks, Relationships, Activity and Audit as relevant.
Main Content	Sections, lines, evidence, relationships and controlled forms.
FactBox	Supporting context, totals, dates, people, warnings and projections.
Sticky Action Area	Save, submit, decision or posting actions according to state.

ID	Area	Requirement	Priority	Acceptance signal
UI-REC-001	Header	All major records use the shared RecordHeader contract.	Must	Code, status, ownership and scope are consistently located.
UI-REC-002	Actions	Show no more than one primary and one secondary visible action plus More.	Must	Action hierarchy remains clear.
UI-REC-003	Lifecycle	State rail shows blockers and required next actor.	Must	Users understand why progress is blocked.
UI-REC-004	Tabs	Tabs are deep-linkable and permission-aware.	Must	Unauthorized tabs are not fetched or displayed.
UI-REC-005	FactBox	Supporting information remains secondary and collapsible.	Should	Primary form remains visually dominant.

16. Record Relationships and Context Drawers

Related records are inspected through a shared relationship drawer or side panel. The drawer shows status and context without duplicating the related record or expanding permission.

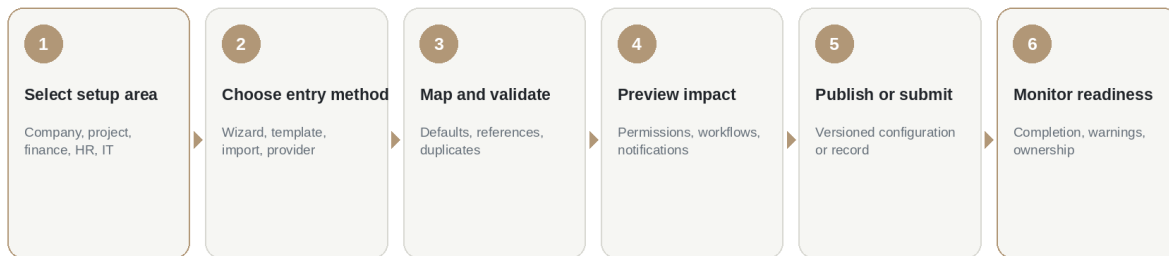
- Relationship type and source module
- Record code, title and status
- Owning company and Project
- Source version and last synchronization where projected
- Permission state and deep link
- Create-linked-action only through the owning module contract

17. Form Architecture

Forms are organized by business meaning rather than by storage model. The platform uses a step, section, group and field hierarchy, with common and required information shown first.

Data Population and Setup Journey

The platform must support guided entry, templates, bulk imports and provider synchronization



Core UX rule: show only the information required for the current step, while preserving full enterprise detail behind progressive disclosure.

Figure 4 - Guided population path for setup, templates, imports and provider connections.

Form principle	Implementation rule
Grid	12-column desktop, 8-column tablet and 4-column mobile layout.
Grouping	Identity, ownership, scope, dates, commercial, technical, people, documents and controls.
Progressive disclosure	Advanced or uncommon fields remain in expandable sections.
Conditional fields	Fields appear only when a controlling value makes them relevant.

Form principle	Implementation rule
Smart defaults	Defaults derive from context, templates, source records and configuration.
Field help	Important unfamiliar fields provide contextual help and examples.
Validation	Field, section and submission-level validation are visible before submission.

ID	Area	Requirement	Priority	Acceptance signal
UI-FRM-001	Required fields	Required fields are visually clear without excessive red markers.	Must	Users can identify unresolved requirements before submit.
UI-FRM-002	Defaults	Inherited and calculated values are identifiable where material.	Must	User can distinguish entered, inherited and calculated values.
UI-FRM-003	Conditionality	Hidden conditional fields are excluded from validation unless active.	Must	Inactive branches do not block submission.
UI-FRM-004	Sensitive fields	Sensitive fields use independent permission-aware components.	Must	Restricted values are never fetched for unauthorized users.
UI-FRM-005	Mobile	Form sections become full-width and touch-friendly.	Must	No mandatory field requires horizontal scrolling.

18. Universal Draft Mode

New → Draft → Validation Ready → Submitted

Draft Mode separates work preservation from business consequence. Saving does not submit, approve, reserve budget, notify external parties or issue a controlled revision.

Draft feature	Behavior
Manual Save	User explicitly saves current draft.
Unsaved state	Header and exit warning indicate unsaved changes.
Draft recovery	User may restore previous workspace or open record only.
Conflict detection	Expected server version is checked when saving.
Draft ownership	Draft owner and permitted collaborators are visible.
Validation	User may run validation without submitting.

19. Guided Wizards

Wizards are required where users cannot reasonably understand all dependencies in a single long form. Every wizard provides purpose, progress, validation, save-and-continue and final review.

Wizard step	Typical content
Basics	Identity, name, type and template.
Scope	Company, Project, location and ownership.
People	Responsible roles and participants.
Business rules	Commercial, technical, lifecycle or configuration choices.
Documents	Required evidence, source revisions and templates.
Review	Impact, unresolved issues and final action.

20. Company Setup Center

Company Setup Center is the primary experience for filling the platform foundation. It presents readiness, ownership, blockers and a direct route into each required configuration area.

Setup area	Minimum content
Company Identity	Legal name, trading name, registration, tax, address, currency, locale and time zone.
Group and Subsidiaries	Parent/child structure, shared-service scope and company relationships.
Branches and Offices	Locations, working time zones and company resources.
Departments and Teams	System departments, hierarchy, managers and team templates.
Users and Roles	Invitations, memberships, role assignments and access readiness.
IT and Integrations	Domains, Microsoft/Google, email, calendar, devices and licences.
Finance Foundations	Fiscal periods, accounts, dimensions, taxes, currencies and approval readiness.
HR Foundations	Positions, schedules, leave, payroll groups and onboarding templates.
Documents	Company folders, numbering, classifications and templates.
Workflows	Approval matrices and published workflow versions.
Launch Review	Unresolved mandatory items, warnings, owner sign-off and activation.

ID	Area	Requirement	Priority	Acceptance signal
UI-CST-001	Readiness	Each setup area shows Not Started, In Progress, Ready, Attention, Optional or Blocked.	Must	Owner can identify launch blockers immediately.
UI-CST-002	Ownership	Every setup item identifies a responsible role or user.	Must	Unowned mandatory setup is flagged.
UI-CST-003	Entry methods	Support manual wizard, template, import and provider connection.	Must	Large companies are not forced into record-by-record entry.
UI-CST-004	Launch	Launch Review validates dependencies before activation.	Must	Critical missing configuration blocks launch.

21. Project Setup Center

Project setup area	Examples
Basic details	Project identity, type, dates, currency, time zone and confidentiality.
Companies and participants	Owner, developer, contractor, consultant, JV and selected external organizations.
Location hierarchy	Site, building, floor, zone, room, unit or infrastructure section.
Project team	Roles, effective dates, work-package and location scope.
Work breakdown	Phases, milestones, packages and activities.
Cost structure	Cost codes, budgets and dimensions.
Document structure	System folders, Tranzit, numbering and templates.
Operational plans	QA/QC, HSE, procurement, reporting and handover.
BIM and Portal	Models, publication profile and external collaboration.

22. Import Center

Upload → Select type → Map → Validate → Detect duplicates → Preview → Resolve → Commit → Reconcile

Import status	Meaning
Ready	Valid row and references.
Warning	Permitted but requires attention.
Duplicate candidate	Potential existing record requires explicit decision.
Invalid	Field or business-rule failure.
Missing reference	Referenced company, Project, user, supplier or code not resolved.
Forbidden scope	Uploader lacks authority for the target context.

ID	Area	Requirement	Priority	Acceptance signal
UI-IMP-001	Formats	Support governed CSV, XLSX, ZIP package and provider imports.	Must	Supported types are visible before upload.
UI-IMP-002	Mapping	Users may save permission-scoped mapping templates.	Should	Repeated imports avoid remapping.
UI-IMP-003	Preview	No authoritative write occurs before preview and authorized commit.	Must	User sees creates, updates and rejects.
UI-IMP-004	Report	Completion produces an immutable import result and downloadable error file.	Must	Every source row has a result.

PART IV - Data, Decision and Collaboration Workspaces

23. Registers and Data Tables

Registers must support enterprise-scale review while remaining visually restrained. Tables use a sticky header, saved views, filter chips, virtualized rendering and safe selection actions.

Capability	Requirement
Columns	Configurable within company and field-permission rules.
Density	Comfortable, Standard and Compact; Standard default.
Sorting	Server-aware sorting with visible active order.
Selection	Selection toolbar shows only safe bulk actions.
Totals	Use only for semantically valid and currency-safe values.
Virtualization	Required for very large datasets.
Mobile	Rows transform into summary cards with primary action.

24. Filters, Views and Personalization

Personalization allowed	Not allowed in initial release
Save filters and sorting.	Replace platform workflow states.
Choose permitted columns.	Change semantic status colors.
Favorite Projects and records.	Create arbitrary custom modules.
Pin authorized records.	Override security or posting rules.
Choose calendar and density preferences.	Freely rearrange fixed Role Center architecture.
Restore previous workspace.	Expose restricted fields through shared views.

25. Approval Workspace

Approvals require a dedicated review surface that keeps evidence and consequences visible. The UI never treats an approval as a generic status change.

Decision panel element	Required information
Decision requested	Approve, reject, return, certify, accept or acknowledge.
Source	Module, record, version and submitted snapshot.
Context	Company, Project, amount, risk, due date and stage.
Evidence	Documents, changes, comparison, comments and previous decisions.
Eligibility	Assigned participant, delegation and separation-of-duty state.
Consequence	What source transition or action follows.
Reason	Required or optional according to workflow definition.

ID	Area	Requirement	Priority	Acceptance signal
UI-APR-001	Desktop	Use side-by-side source/evidence and decision panel where space permits.	Must	Reviewer can compare without losing decision context.
UI-APR-002	Mobile	Use sequential review with fixed final actions after evidence.	Must	Decision cannot be made before required review state.
UI-APR-003	Version	Show exact source version and detect material change.	Must	Stale approval is blocked.

ID	Area	Requirement	Priority	Acceptance signal
UI-APR-004	Reauthentication	High-risk decisions invoke recent authentication/MFA.	Must	Identity confirmation does not bypass missing authority.

26. Transaction and Journal Workspace

Desktop behavior	Mobile behavior
Dense line grid with validation sidebar.	Guided line-entry sequence.
Sticky totals and balance.	Review lines and totals before submit/post.
Source document preview.	Open source evidence in full-screen viewer.
Batch and line-level errors.	Errors grouped by step and line.
Server-confirmed posting.	Posting requires online confirmation.

27. Activity, Comments and Timeline

Every major record uses one timeline language for user-facing activity. Audit remains a separate governance view.

Activity category	Examples
Lifecycle	Submitted, approved, issued, closed, reopened.
Collaboration	Comment, mention, clarification and assignment.
Document	Revision attached, published or superseded.
Integration	External mapping, sync or provider result.
Correction	Reversal, adjustment, superseding revision or conflict resolution.
System	Automated deadline, reminder, reconciliation or projection update.

28. Documents UI

Component	Behavior
Document Picker	Link existing logical document, current revision, exact revision or upload new draft.
Viewer	Metadata, revision state, file preview, relationships and access policy.
Revision history	Clearly distinguish current, superseded and draft revisions.
Duplicate detection	Offer link existing, add revision or continue separately according to policy.
Security	View, download, watermark and signed URL follow document and source permissions.

29. Global Search

Search is a primary navigation surface. It must apply tenant, company, Project, record and field permissions before returning title, snippet, count or autocomplete information.

Result element	Required content
Identity	Type, title/code and status.
Context	Company, Project and primary relationship.
Match	Permission-safe matched text or metadata.
Access state	Open, locked minimal result or unavailable.
Action	Open, request access or view related source.

30. Notifications and Event Center

Section	Purpose
Needs Action	Decision, acknowledgement, clarification or required work.
Important	High-impact updates without direct action.
Mentions	Comments and direct references.
Updates	Grouped operational changes.
System	Security, integration, sync or platform notices.

Notification grouping should reduce noise. A record may also create a Work Inbox item, banner or record warning, but each surface must reflect the same authoritative event and policy.

31. Status and Lifecycle Design

Semantic family	Visual treatment	Examples
Neutral	Text and subtle neutral chip	Not started, inactive, informational.
Draft	Neutral/gold-soft chip	Draft, prepared, local draft.
In progress	Restrained information treatment	Under review, active, processing.
Success	Restrained green	Approved, passed, completed.
Warning	Restrained amber	Expiring, attention, partial.
Critical	Restrained red	Failed, overdue critical, security risk.
Blocked	Strong icon and reason	Legal hold, dependency, permission block.
Archived	Muted chip	Archived, superseded, retired.

32. Validation, Errors and Concurrency

Error type	UI behavior
Field validation	Inline message beside the field.
Section validation	Section heading shows issue count and focuses first issue.
Submission block	Summary explains every unresolved blocker.
Permission error	Explain missing scope or role without exposing restricted content.
Business-rule block	Explain governing rule and next permitted action.
Version conflict	Side-by-side change review; never silent overwrite.
Provider failure	Show provider, retryability, last success and support route.
Offline restriction	Preserve draft and explain online requirement.

33. Required UI States

- Loading and geometry-matched skeleton
- Empty but authorized
- No access or no relationship
- Configuration disabled
- Draft, submitted and locked
- Archived or superseded
- Offline, pending, synchronizing, failed and conflicted
- Permission revoked while open
- Partial or stale projection
- Service temporarily unavailable
- File quarantined or processing

34. Empty States

Empty states explain the purpose of the area, the reason it is empty and the next valid action. Users without create permission receive a scope or access explanation instead of a misleading create button.

ID	Area	Requirement	Priority	Acceptance signal
UI-EMP-001	Actionability	Authorized empty states include relevant create, import or learn actions.	Must	User understands how to populate the area.
UI-EMP-002	Permission	Unauthorized empty states do not imply no data exists globally.	Must	No record-existence leakage.
UI-EMP-003	Visual restraint	Use concise text and minimal illustration.	Should	Empty states do not dominate the workspace.

35. Contextual Help

Help content	Requirement
Meaning	Plain-language definition.
Why it matters	Business consequence or reporting use.
Responsibility	Who supplies or approves the value.
Format	Expected type, unit or code.
Example	Realistic non-sensitive example.
Next step	What happens after completion.
Learn more	Link to governed internal documentation where necessary.

PART V - Responsive, Mobile, Offline and Specialized Experiences

36. Responsive Architecture

Component	Desktop	Tablet	Mobile
Role Center	Multi-column high-value zones.	Two-column adaptive zones.	Single-column priority stack.
Register	Full table with FactBox.	Simplified table or split view.	Card list plus filter sheet.
Record	Header, tabs, FactBox.	Collapsible two-column.	Header and segmented tabs; drawers.
Approval	Side-by-side review.	Adaptive split.	Sequential review then decision.
Transaction	Dense line grid.	Simplified grid.	Guided line-entry flow.
Configuration	Tree and detail.	Collapsible panels.	Focused settings flows only.

37. Mobile and Field Experience

```

Field Home
├── My Work
│   ├── Inspections
│   ├── Tasks
│   ├── HSE
│   ├── Daily Reports
│   ├── Deliveries
│   ├── Assets
│   └── Sync Queue

```

- Scan-first navigation for asset, unit, location and document QR/barcodes.
- Camera-first evidence capture and attachment queue.
- Large pass/fail, measurement, stepper and signature controls.
- Location breadcrumb and Project context visible during capture.
- Minimal typing with templates, defaults and recent permitted values.
- No decorative motion or blocking animation in field workflows.

38. Offline Experience

Offline state	User meaning
Local Draft	Exists only on the device.
Pending Upload	Queued and waiting for connection/dependency.
Synchronizing	Upload or command processing in progress.
Synchronized	Server accepted and returned authoritative result.
Failed	Retryable or permanent technical failure.
Conflict	Server record changed; human or rule-based resolution required.
Rejected	Server rejected due to permission or business rule.

ID	Area	Requirement	Priority	Acceptance signal
UI-OFF-001	Global status	Persistent offline bar shows queued action count.	Must	User never assumes offline data was delivered.
UI-OFF-002	Queue	Sync Queue lists record, action, attachments, dependency and result.	Must	Every local command is traceable.
UI-OFF-003	Revocation	Permission changes preserve local draft but block unauthorized submission.	Must	Protected cache follows expiry policy.
UI-OFF-004	Conflict	Conflicts are visible and never resolved by silent last-write-wins.	Must	User can review source/current differences.

39. BIM and Spatial Experience

Viewer region	Purpose
Model tree	Discipline, layer, classification and object navigation.
Search and filters	Find object, space, asset, unit or operational status.
Viewer canvas	3D model, floor plan, map or split view.
Object panel	Properties, stable object identity and linked operational records.
Viewpoints/markups	Controlled spatial comments and issue context.
Operational overlays	Progress, quality, tasks, assets or document state.
Timeline	Version and linked-record change history.

40. Client and Supplier Portals

Portal	Primary navigation
Client	Home, Project, Unit, Progress, Documents, Financial Summary, Meetings, Requests, Handover, Defects and Support.
Supplier	Home, Profile, RFQs, Quotations, Purchase Orders, Deliveries, Submittals, Timesheets, Invoices, QA/QC, HSE and Documents.

- Use lower information density and externally understandable terminology.
- Every external write is shown as a submission, request or acknowledgement.
- Organization branding may affect logo and approved accents without changing component behavior.
- Permission and relationship expiry states are explicit.
- Mobile priorities are specific to the client or supplier workflow.

PART VI - Configuration, Permissions and Design System

41. Platform Configuration UI

Platform Default
 → Tenant
 → Business Group
 → Company
 → Project
 → Module
 → Record Type

Configuration area	Examples
Navigation	Module availability, ordering and labels.
Pages and sections	Visible tabs, sections and widgets.
Fields	Visibility, required rules, help, order and options.
Statuses	Allowed configured labels and transitions within domain rules.
Templates	Company, Project, record, document and workflow templates.
Numbering	Record code formats and sequences.
Permissions	Roles, capabilities, fields and scope.
Notifications	Policies, channels and templates.
Reports	Enabled reports, saved filters and distribution.
Integrations	Connections, mappings and synchronization rules.

ID	Area	Requirement	Priority	Acceptance signal
UI-CFG-001	Versioning	Configuration changes are draft, validated, previewed and published as versions.	Must	Historical record meaning remains stable.
UI-CFG-002	Preview	Admin may preview as selected roles and surfaces.	Must	Visibility and actions are tested before publish.
UI-CFG-003	Inheritance	UI shows inherited value and override source.	Must	Admin understands why a setting is effective.
UI-CFG-004	Rollback	Published configuration can be rolled back safely.	Should	Rollback preserves audit and effective dates.

42. Permission-Aware Interface

Unavailable state	Preferred UI treatment
Ordinary inaccessible action	Hide.
Useful explanatory restriction	Show disabled with reason.
Record visible, field restricted	Hide, redact, mask or summarize according to field policy.
Workflow blocked	Show current stage and required participant.
Offline restricted	Show online requirement and preserve draft.
Configuration disabled	Remove entry point and preserve historical read where permitted.
Permission revoked while open	Immediately lock sensitive content and explain scope change.

43. Design System Foundations

The ConstructionOS design language should feel architectural, precise and premium. It uses light surfaces, dark-grey typography, restrained company or gold accents, clear spacing and minimal shadow.

Use	Avoid
Clear hierarchy and controlled density.	Oversized cards that reduce information efficiency.
Borders and spacing to separate structure.	Heavy shadows and glass effects.
Restrained accent for selected and premium states.	Gold on every button or status.
Functional diagrams and icons.	Playful SaaS illustrations.
Consistent geometry and alignment.	Different radius and spacing systems per module.
Calm micro-interactions.	Bouncing, particles, parallax or decorative delays.

44. Visual Tokens: Color, Type, Spacing and Shape

Token	Recommended baseline	Use
Canvas	#F6F6F3	Application background.
Surface	#FFFFFF	Cards, records and overlays.
Surface Secondary	#F0F0ED	Secondary regions and grouped rows.
Border	#DEDED8	Structural separators.
Text Primary	#242422	Primary typography.
Text Secondary	#666662	Supporting content.
Premium Accent	#B19777	Selected, premium and refined emphasis.
Accent Soft	#EEE7DE	Subtle selected surface.

Typography role	Size baseline
Display	32 px
Page title	24 px
Section heading	18 px
Card title	15 px
Body	14 px
Compact data	13 px
Metadata	12 px

Geometry	Baseline
Spacing scale	4 / 8 / 12 / 16 / 24 / 32 / 48 px
Input and button radius	8 px
Card radius	12 px
Dialog radius	16 px
Status chip	Pill / full radius
Shadow	Reserved for menus, drawers, dialogs and drag state

45. Buttons, Actions and Confirmation

Action level	Examples	Rule
Primary	Save, Submit, Approve, Post, Issue	One primary action per active region.
Secondary	Save Draft, Validate, Preview, Return	Supports the main path.
Tertiary	Cancel, open related, show history	Low emphasis.
Destructive	Delete, revoke, reverse, remove access	Never use ordinary primary styling.

High-risk confirmations must state the record, company/Project context, consequence and irreversibility. Very high-risk actions may require reauthentication, reason, typed confirmation and workflow approval.

46. Drawers, Dialogs and Full-Page Editors

Container	Use for	Do not use for
Drawer	Preview, related records, filters, activity and low-complexity draft entry.	Large multi-section or high-risk forms.
Dialog	Confirmation, one focused decision or short selection.	Long editing workflows.
Full page	Complex records, formal submissions, approvals, transactions and configuration.	Small reversible choices.
Bottom sheet	Mobile filters, actions, help and compact selection.	Long document review.

47. Accessibility and Localization

- WCAG 2.1 AA baseline for core workflows.
- Keyboard navigation and visible focus for all interactive controls.
- Screen-reader labels, semantic headings and accessible table relationships.
- No status, error or validation communicated only by color.
- Reduced-motion support and no mandatory motion to understand state.
- Minimum touch-target sizing and mobile text scaling.
- Locale-aware dates, numbers, currencies, time zones and units.
- English and Albanian resource architecture, with right-to-left readiness for future expansion.
- Original-language preservation for issued documents and external submissions.

48. Motion and Loading

Interaction	Timing baseline
Micro interaction	120-160 ms
Panel / drawer	180-220 ms
Page transition	200-250 ms
Status confirmation	180-240 ms

- Skeletons match final component geometry.
- Avoid full-page spinners and layout shift.
- Long operations expose current step, progress, cancellability and background behavior.
- Motion is interruptible and disabled or simplified under reduced-motion preference.

49. Navigation Memory and Workspace Restoration

ConstructionOS may remember active company, Project, tab, filters, table columns, scroll position, open context panel and draft. On return, the user chooses whether to restore the previous workspace.

ID	Area	Requirement	Priority	Acceptance signal
UI-MEM-001	Scope	Restoration revalidates membership and permissions.	Must	Revoked context is not restored.
UI-MEM-002	Draft	Unsaved local draft is clearly distinguished from server draft.	Must	User understands which version is authoritative.
UI-MEM-003	Privacy	Sensitive content is not restored after logout or session expiry without reauthentication.	Must	Session policy is enforced.

PART VII - Shared Components and Frontend Engineering

50. Component Library

Component family	Components
Foundation	Button, icon button, input, textarea, select, autocomplete, date/time, checkbox, radio, switch, file input, badge, tooltip and popover.
Layout	AppShell, Sidebar, TopBar, PageHeader, RecordHeader, Section, Card, Grid, SplitView, FactBox, Drawer, Dialog and StickyActionBar.
Data	DataTable, CardList, FilterBar, SavedViewSelector, KPI, ChartContainer, Timeline, ActivityFeed, RelationshipList, DocumentList and AuditList.
Workflow	StatusRail, ApprovalPanel, DecisionCard, ValidationSummary, ConflictResolver and WorkflowHistory.
Domain fields	PersonPicker, CompanyPicker, ProjectPicker, LocationPicker, SupplierPicker, DocumentPicker, CurrencyField, QuantityField, ConfidentialField and EffectiveDateRange.

ID	Area	Requirement	Priority	Acceptance signal
UI-CMP-001	Contract	Every shared component defines states, accessibility, responsive and permission behavior.	Must	Story/test coverage exists for the component contract.
UI-CMP-002	Versioning	Components and tokens are versioned with migration guidance.	Must	Module upgrades do not silently change critical behavior.
UI-CMP-003	Exceptions	One-off components require documented review and reusable rationale.	Must	Design-system drift is controlled.

51. Schema-Driven Forms

```
Field Definition
├── Type
├── Label and help
├── Required rule
├── Visibility rule
├── Validation
├── Permission
├── Default / source
├── Layout
└── Configuration version
```

Schema-driven forms are recommended for configurable company/project fields, localized labels and consistent responsive behavior. Core financial, legal, identity and lifecycle controls remain strongly typed platform components.

52. Frontend Technical Architecture

Frontend Technical Architecture

Shared platform capabilities keep every module visually and behaviorally consistent

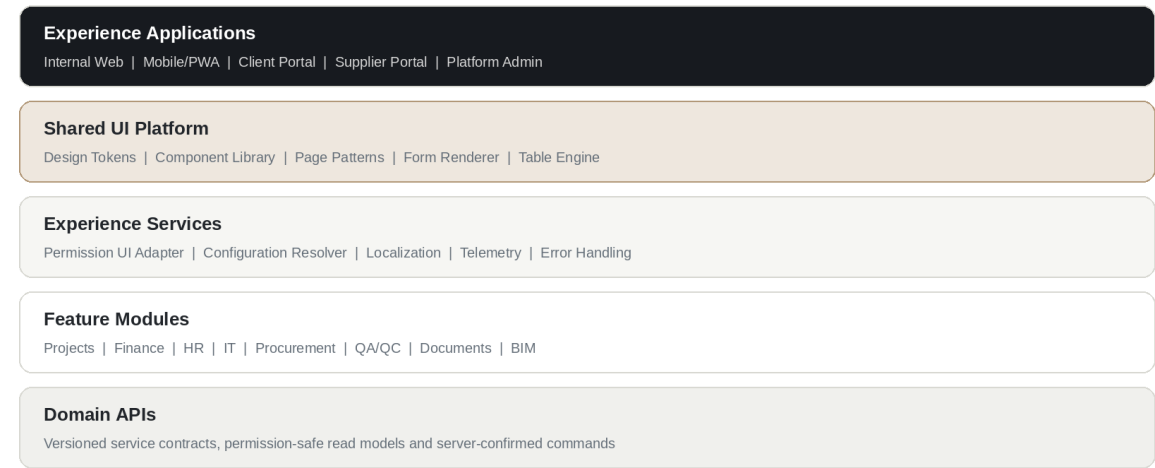


Figure 5 - Shared frontend platform and feature-module layering.

Layer	Responsibility
Experience applications	Channel-specific shells and navigation.
Shared UI Platform	Tokens, components, page patterns, forms, tables and experience services.
Feature modules	Domain-specific pages assembled from shared contracts.
BFF / APIs	Permission-safe read models and versioned commands.
Domain services	Authoritative business state and lifecycle.

53. Routes, State and Data Fetching

Area	Rule
Routes	Feature routes are deep-linkable and permission-guarded.
URL state	Filters, sorting, tabs and selected context are URL-addressable where safe.
Server state	Authoritative records, permissions, workflow and timelines remain server-owned.
Client state	UI preferences, local drafts and safe optimistic intent only.
Caching	Tenant/company/scope-aware keys and revocation-aware invalidation.
Optimism	Allowed only for reversible low-risk actions.
Conflicts	Expected version and explicit conflict UI.

```
/dashboard
/work
/search
/projects
/projects/:projectId
/projects/:projectId/setup
/finance/invoices/:invoiceId
/hr/employees/:employeeId
/it/integrations/:connectionId
/settings/:area
```

54. Security and Privacy

- Frontend hiding is not authorization; backend queries and read models enforce row and field security.
- Active tenant, company, Project and role context is attached to protected requests.
- Search, exports, notifications, caches and analytics use the same permission policy.
- Sensitive reads may trigger audit according to classification and configuration.
- Session, impersonation, support and break-glass context is persistently visible.
- File previews and downloads revalidate access before issuing signed URLs.
- Browser storage contains no unrestricted authoritative business data.

55. Analytics and UX Telemetry

Metric	Use
Time to complete record	Identify unnecessary form complexity.
Validation error frequency	Improve help and defaults.
Form abandonment	Identify confusing or blocked workflows.
Search success	Improve ranking and terminology.
Navigation depth	Reduce excessive routing.
Help usage	Identify unclear concepts.
Mobile sync failure	Improve field reliability.
Decision time	Improve approval evidence and routing.
Page performance	Maintain responsive experience.

Analytics guardrail

Interface telemetry may improve the product and process. It must not automatically be presented as an individual employee performance score.

56. Non-Functional Requirements

Area	Requirement
Performance	Typical authenticated page initial load target under 2.5 seconds on supported broadband; common API reads p95 under 500 ms excluding heavy files/reports.
Search	Permission-filtered common search target p95 under 2 seconds.
Scalability	Virtualized registers and horizontally scalable web/API architecture.
Reliability	Draft recovery, idempotent commands and observable background work.
Browsers	Current and previous major versions of Chrome, Edge, Safari and Firefox.
Accessibility	WCAG 2.1 AA target for core workflows.
Localization	English and Albanian resources; locale formatting separated from stored values.
Observability	Structured logs, metrics, traces and tenant-safe diagnostics.

PART VIII - Quality, Delivery and Governance

57. Testing Strategy

Test layer	Coverage
Design-system unit	Component states, keyboard, screen reader, responsive and token use.
Visual regression	Shared components, page patterns and critical module pages.
Permission tests	Hidden/disabled/locked/field-redacted behavior and backend response shaping.
Responsive tests	Desktop, tablet and mobile breakpoints with content stress.
Accessibility tests	Automated audit plus manual keyboard and assistive-technology review.
Workflow tests	Draft, validate, submit, approve, return, post and issue.
Conflict tests	Concurrent edits, revoked access, stale version and offline conflict.
Performance tests	Large tables, complex forms, document previews and BIM-linked views.
Localization tests	Long translations, locale formats and original-language preservation.

58. Acceptance Criteria

ID	Acceptance criterion	Priority
UI-AC-001	Every role receives a focused Role Center.	Must
UI-AC-002	Users do not see irrelevant modules or ordinary inaccessible actions by default.	Must
UI-AC-003	Active company and Project context are always clear.	Must
UI-AC-004	Every major record uses the shared record-page architecture.	Must
UI-AC-005	Save, Submit, Approve, Post, Issue and More remain consistently positioned.	Must
UI-AC-006	Complex records use Universal Draft Mode.	Must
UI-AC-007	Company and Project Setup Centers guide initial population.	Must
UI-AC-008	Imports support mapping, preview, duplicate handling and result reporting.	Must
UI-AC-009	Forms use progressive disclosure and contextual help.	Must
UI-AC-010	Tables convert to usable mobile cards.	Must
UI-AC-011	All pages define loading, empty, error, permission, offline and conflict states.	Must
UI-AC-012	Search, exports and previews respect row- and field-level permissions.	Must
UI-AC-013	Offline status and Sync Queue are understandable.	Must
UI-AC-014	Version conflicts never overwrite silently.	Must
UI-AC-015	Disabled configuration disappears cleanly while history is preserved.	Must
UI-AC-016	Portals use simplified external terminology and published projections only.	Must
UI-AC-017	BIM and spatial workspaces use the shared shell and relationship language.	Must
UI-AC-018	Motion feels refined without delaying work.	Must
UI-AC-019	Accessibility baseline is met for core workflows.	Must

ID	Acceptance criterion	Priority
UI-AC-020	New modules are assembled from shared components and page patterns.	Must

59. Delivery Phases

Phase	Scope	Exit condition
1 - Foundations	Tokens, typography, color, spacing, accessibility and basic components.	Design-system package and documentation approved.
2 - Shell and navigation	AppShell, context, navigation, Role Center, Work Inbox, search and notifications.	Core company workspace navigable by representative roles.
3 - Record and forms	Record page, forms, Draft Mode, wizards, tables, filters and relationships.	Projects, Tasks and Documents demonstrate shared patterns.
4 - Setup and imports	Company/Project Setup Centers, templates, imports and configuration preview.	New company can reach launch readiness.
5 - Decisions and transactions	Approval and journal workspaces, conflict handling and reauthentication.	Finance/Procurement/Workflow critical flows pass.
6 - Mobile and portals	Field shell, offline states, Client and Supplier Portals.	Critical responsive and offline flows certified.
7 - BIM and advanced governance	Spatial workspace, schema-driven forms, visual regression and telemetry.	Platform UI baseline signed off.

60. Page and Flow Registry

Code	Page / flow	Pattern	Primary purpose
UI-001	Role Center	Dashboard	Role-specific priorities and performance.
UI-002	Work Inbox	Register	Actionable tasks, approvals and exceptions.
UI-003	Global Search	Search	Permission-safe cross-module navigation.
UI-004	Company Setup	Wizard / setup	Populate company foundation.
UI-005	Project Setup	Wizard / setup	Populate project foundation.
UI-006	Import Center	Wizard / register	Bulk population and reconciliation.
UI-007	Universal Register	Register	Filter, review, bulk act and export.
UI-008	Universal Record	Record	Authoritative record workspace.
UI-009	Universal Form	Form	Create/edit draft.
UI-010	Approval Workspace	Approval	Evidence review and decision.
UI-011	Transaction Workspace	Transaction	Line-based controlled entry.
UI-012	Document Viewer	Viewer	Revision-aware controlled file access.
UI-013	Activity / Timeline	Timeline	User-facing history and collaboration.
UI-014	Configuration Editor	Configuration	Versioned rules and preview.
UI-015	Field Home	Mobile dashboard	Assigned site work and capture.
UI-016	Sync Queue	Mobile register	Offline command and upload state.
UI-017	BIM Viewer	Spatial	Model and linked operational context.
UI-018	Client Portal Home	Portal dashboard	Published client work.
UI-019	Supplier Portal Home	Portal dashboard	Supplier submissions and status.
UI-020	Platform Admin Shell	Admin	Tenant-safe SaaS operations.

61. Risks and Open Decisions

Topic	Decision / risk to manage
Frontend framework	Confirm implementation framework and shared-package strategy.
Design-token delivery	Confirm token format, theming boundaries and company-brand override limits.
Schema-driven forms	Define which fields remain strongly typed and which are metadata-rendered.
Role Center configuration	Confirm exact company-configurable regions without arbitrary layout fragmentation.
Locked search results	Define which record types may expose a minimal locked result.
Offline scope	Confirm first-release record types and protected-cache expiry.
BIM viewer	Confirm viewer technology and mobile fallback.
Portal branding	Define approved brand customization and accessibility constraints.
Keyboard shortcuts	Remain out of V1 unless separately approved; accessibility keyboard operation still required.
Dark mode	Not part of confirmed baseline; future evaluation only.

62. Definition of Done

- All Must requirements and acceptance criteria are implemented and tested.
- The design-token package, component library and page-pattern library are versioned and documented.
- Representative modules demonstrate Role Center, register, record, form, approval, transaction and configuration patterns.
- Company and Project setup journeys can populate a new environment without hidden mandatory steps.
- Permissions, configuration, search, export, portals and mobile are tested for cross-company and field-level leakage.
- Responsive, accessibility, localization and reduced-motion reviews pass.
- Visual regression coverage protects shared components and critical pages.
- Performance targets are met with production-scale representative data.
- No unresolved critical security, accessibility, data-loss or interface-consistency defect remains.

Final product position

ConstructionOS must feel like one premium company operating system with role-specific workspaces, guided data entry, predictable records, powerful registers, controlled workflows and a consistent visual language.

END OF DOCUMENT